Akshat Ramachandran

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Research Overview

Vision

Design the next generation of AI inference infrastructure by treating efficiency as a first-class design objective. My work focuses on cross-layer co-design across model compression algorithms, runtime systems, and specialized hardware architectures to ensure the future of AI scales through efficiency, not just compute.

Interests

Computer Architecture, Efficient AI, Hardware-Software Co-Design, Foundation Models (LLMs and VLMs), Computer Vision, Computer Arithmetic, Digital Circuit Design

Languages and Tools

Python, C/C++, Verilog, SystemVerilog, HLS (Vivado, Catapult), CUDA, Triton, ML frameworks (PyTorch, Keras), HW design tools (Synopsys Design Compiler, Innovus)

Education

Georgia Institute of Technology — Aug 2023 – May 2027
Ph.D. in Electrical and Computer Engineering **| Atlanta, GA** — GPA: 4.0/4.0
Minor: Finance (Scheller College of Business)
Thesis: Co-Designing Hardware and Algorithms for Efficient Foundation Model Computing via Adaptable Sparsity and Quantization
Advisor: Dr. Tushar Krishna

Georgia Institute of Technology — Aug 2023 – Dec 2025
M.S. (Thesis) in Electrical and Computer Engineering **| Atlanta, GA** — GPA: 4.0/4.0
Thesis: Algorithm-Hardware Co-Design of Digital Compute-in-Memory Architecture Supporting Flexible and Temporal N:M Sparsity
Advisor: Dr. Tushar Krishna

Veermata Jijabai Technological Institute (VJTI) — Aug 2018 – May 2022
B.Tech. in Electronics Engineering **| Mumbai, India** — GPA: 9.80/10.0
Department Rank: 1/83 (Gold Medalist) | Institute Rank: 2/480
Thesis: Next Generation Architecture for Computer Vision
Advisor: Dr. John Gustafson (Arizona State University)

Professional Experience

Infravana — Oct 2026 – Present
Founding Engineer Intern, Seed-stage startup | San Francisco, CA

- Building an LLM serving layer across heterogeneous compute infrastructure (GPUs, TPUs, etc.) to improve cost efficiency while meeting latency SLOs.

NVIDIA Research — May 2026 – Present
Ph.D. Research Intern, Accelerators and VLSI (Manager: Dr. Brucek Khailany) | Santa Clara, CA

- Developed a runtime-adaptive quantization paradigm that adjusts weight, activation, and KV cache precision dynamically during long-horizon agentic inference.
- Designed a composable, multi-scale quantization format built on a shared NVFP4 representation for runtime precision adaptation.
- Optimized the Tensor Core datapath with a 2-bit quinary format, achieving 2× baseline NVFP4 prefill throughput.

NVIDIA Research May 2025 – Aug 2025
Ph.D. Research Intern, Accelerators and VLSI (Manager: Dr. Brucek Khailany) | Santa Clara, CA

- Proposed ThinKV, a thought-adaptive KV-cache compression technique that exploits the heterogeneous importance of reasoning tokens to enable selective quantization and token eviction for efficient long-context reasoning.
- Achieved near-lossless accuracy while reducing KV-cache memory usage by more than 96%.
- Filed a patent and led work resulting in an ICLR 2026 oral presentation.

Intel Labs Aug 2024 – Dec 2024
AI Hardware Research Intern (Managers: Dr. Arnab Raha, Dr. Souvik Kundu) | Santa Clara, CA

- Proposed a layer-wise outlier-density-aware N:M sparsity selection method and FlexCiM, a digital compute-in-memory architecture with flexible structured sparsity support.
- Integrated FlexCiM into Intel NPU-v7, resulting in an ISLPED 2025 Best Paper Award and a patent filing.

NVIDIA Corporation May 2024 – Aug 2024
Architecture Energy Modeling Intern (Manager: Dr. Visu Subramanian) | Santa Clara, CA

- Developed an AI-based power estimation tool for NVIDIA GPUs.
- Leveraged transfer learning to distill knowledge from parent chips to derivative architectures, reducing data collection and training effort by 5×.

Georgia Institute of Technology Aug 2023 – Present
Graduate Research Assistant (Advisor: Dr. Tushar Krishna) | Atlanta, GA

- Conducting doctoral research on co-designing hardware and algorithms for efficient foundation model computing via adaptable sparsity and quantization.

Lemurian Labs Jun 2023 – Aug 2023
Hardware Intern (Managers: Dr. Ashish Kaul, Dr. Theodore Omtzigt) | Menlo Park, CA

- Designed and verified a smart NoC to interlink functional units in the Lemurian tensor engine using SystemVerilog and Python/PyUVM, achieving a 35% increase in operating frequency and 2.5× better performance over the existing interconnect.
- Spearheaded deployment of an adaptive logarithmic datatype in Python and C++, and evaluated mixed-precision quantization of NumPy-implemented BERT and GPT, yielding a 10× speedup over floats.

Samsung Research & Development Aug 2022 – Jul 2023
Visual Intelligence Engineer (Managers: Dr. B.V. Vandrotti, Mr. Jooyoung Kim) | Suwon, South Korea

- Contributed to accelerator and algorithm co-design, optimization, and research-to-commercialization of depth perception and processing solutions for XR devices, leading to an ICIP publication, a patent, and a Samsung Best Paper Award nomination.
- Co-designed and cross-validated a deep-learning-based ToF depth fusion algorithm and SoC architecture in Python and HLS, enabling a 67% reduction in on-device inference latency.
- Designed a systolic dataflow accelerator in SystemVerilog for ToF acquisition, reducing latency by 60%.

Indian Institute of Technology Bombay Jul 2021 – Dec 2022
Research Assistant (Advisor: Dr. Madhav Desai) | Mumbai, India

- Designed and verified a reconfigurable, programmable, multithreaded 40 Gb/s NIC and MAC architecture in HLS and Verilog, interfaced as a memory-mapped peripheral to the 2 × 4 × 32 Ajit processor in a MeiTY-funded project.
- Implemented custom packet generation and forwarding flows for verification and NIC support libraries in C.

Samsung Research & Development May 2021 – Jul 2021
Summer Research Intern (Manager: Dr. Ankit Dhiman) | Bangalore, India and Suwon, South Korea

- Researched memory-efficient deep-learning algorithms and architectures for AR applications.
- Remodeled Samsung's 3D reconstruction pipeline in Python and C, and designed a memory controller that extended k-means-style clustering to parameters, achieving nearly 2× higher performance than prior pipelines.

Select Tapeouts and Projects

Tapeout: Unified Prefill-Decode Disaggregation Chip In Progress
Samsung 14 nm FinFET | Role: Algorithm-to-GDS

- Designing a unified DCiM (digital compute-in-memory) and DPU (dot product unit) with unified SRAM to support LLM prefill-decode disaggregation on a single chip.

Project: Hierarchical Mixed-Precision Multi-Agent Systems In Progress

- Developing a hierarchical multi-agent runtime that uses task dependencies, handoff requirements, and confidence to guide mixed-precision sub-LLM spawning and execution under resource budgets.
- The mixed-precision scheme mitigates quantization-induced unnecessary and incorrect tool calls, minimizing the inference overhead of the ReAct loop.

Tapeout: GNN and Transformer Accelerator SoC Jan 2024 – Nov 2024
TSMC 28 nm | Role: Sparse Accelerator RTL

- Contributed to the design of a sparse PE in the SoC that can support mixed N:M sparsity.

Publications (* Equal Contribution)

Conference Publications

Polestar: Drift-Aware Cache Calibration and Token Commitment for Efficient Inference of Diffusion LLMs
Mingyu Lee*, **Akshat Ramachandran***, Souvik Kundu, and Tushar Krishna
*Conference on Neural Information Processing Systems (**NeurIPS**), Dec 2026*

ThinKV: Thought-Adaptive KV Cache Compression for Efficient Reasoning Models
Akshat Ramachandran, Marina Neseem, Charbel Sakr, Rangharajan Venkatesan, Brucek Khailany, and Tushar Krishna
*International Conference on Learning Representations (**ICLR**), Apr 2026*
Oral presentation (Top 1.13% of accepted papers)

OuroMamba: A Data-Free Quantization Framework for Vision Mamba
Akshat Ramachandran*, Mingyu Lee*, Huan Xu, Souvik Kundu, and Tushar Krishna
*International Conference on Computer Vision (**ICCV**), Oct 2025*

Accelerating LLM Inference with Flexible N:M Sparsity via a Fully Digital Compute-in-Memory Accelerator
Akshat Ramachandran, Souvik Kundu, Arnab Raha, Shamik Kundu, Deepak A. Mathaikutty, and Tushar Krishna
*International Symposium on Low Power Electronic Design (**ISLPED**), Aug 2025*
Best Paper Award

MicroScopiQ: Accelerating Foundational Models through Outlier-Aware Microscaling Quantization
Akshat Ramachandran, Souvik Kundu, and Tushar Krishna
*International Symposium on Computer Architecture (**ISCA**), Jun 2025*

AIRCHITECT v2: Learning the Hardware Accelerator Design Space through Unified Repre-

sentations
Jamin Seo*, **Akshat Ramachandran***, Yu-Chuan Chuang, Anirudh Itagi, and Tushar Krishna
*Design, Automation and Test in Europe Conference (**DATE**), Mar 2025*

CLAMP-ViT: Contrastive Data-Free Learning for Adaptive Post-Training Quantization of ViTs
Akshat Ramachandran, Souvik Kundu, and Tushar Krishna
*European Conference on Computer Vision (**ECCV**), Oct 2024*

Algorithm-Hardware Co-Design of Distribution-Aware Logarithmic-Posit Encodings for Efficient DNN Inference
Akshat Ramachandran, Zishen Wan, Geonhwa Jeong, John Gustafson, and Tushar Krishna
*Design Automation Conference (**DAC**), Jun 2024*
Acceptance rate: 19%

PositIV: A Configurable Posit Processor Architecture for Image and Video Processing
Akshat Ramachandran, John Gustafson, Anusua Roy, Rizwan Ahmed Ansari, and Rohin Daruwala
*Euromicro Conference on Digital System Design (**DSD**), Aug 2023*

Journal Publications

GPU Based Building Footprint Identification Utilising Self-Attention Multiresolution Analysis
Rizwan Ahmed Ansari*, **Akshat Ramachandran***, and Winnie Thomas
All Earth, 2023

Workshop Publications

Polestar-Cache: Reconciling Parallel Decoding and Accuracy in Diffusion LLMs via Token Drift-Aware KV Cache Recalibration
Mingyu Lee*, **Akshat Ramachandran***, Souvik Kundu, and Tushar Krishna
Workshop on Latent & Implicit Thinking – Going Beyond CoT Reasoning @ ICLR 2026

Recap: Training-Free Compensation for Coarse Activation Channel Pruning in Compressed LLMs
Mingyu Lee*, **Akshat Ramachandran***, and Tushar Krishna
Machine Learning for Computer Architecture and Systems @ ISCA 2025

Preprints

OneDSE: A Unified Microprocessor Metric Prediction and Design Space Exploration Framework
Ritik Raj, **Akshat Ramachandran**, Jeff Nye, Shashank Nemawarkar, and Tushar Krishna
arXiv

Patents

Think Before you Quantize: Thought-Aligned Adaptive Quantization via Algorithm-Hardware Co-Design for Efficient Large Reasoning Models
Akshat Ramachandran, Marina Neseem, Charbel Sakr, Rangharajan Venkatesan, and Brucek Khailany
Provisional patent filed based on work at NVIDIA Research

Compute-in-Memory Architecture for Accelerating Neural Network Operation
Akshat Ramachandran, Arnab Raha, Souvik Kundu, Shamik Kundu, and Deepak Mathaikutty
US patent filed based on work at Intel Corporation

A Method and System for Acquiring, Processing and Monocular Depth Completion/Estimation of Sparse Time-of-Flight Sensors
Akshat Ramachandran, Ankit Dhiman, Basavaraja Shanthappa Vandrotti, Jooyoung Kim, and Lokesh Boregowda
US Patent, Oct 2024 (US20240331176A1), based on work done at Samsung Research & Development

Awards and Honors

MLCommons Rising Star, Cohort featuring emerging leaders of ML and systems research 2026

Best Paper Award, ACM/IEEE International Symposium on Low Power Electronic Design (ISLPED) 2025

2nd Place, Student Research Competition, IEEE/ACM International Symposium on Microarchitecture (MICRO) 2024

DAC Young Fellow, ACM/IEEE Design Automation Conference (DAC) 2024

Samsung Best Paper Award Nominee, Top 5% of advanced research at Samsung 2023

Samsung Spot Award, For contributions to AR/VR algorithms on Samsung Galaxy devices 2022

Best Undergraduate Thesis Award 2022

Professional Level SW Competency, Global Samsung Aptitude Test 2022

2nd Place, Electrothon Hackathon 2022

2nd Place, Vision Beyond Limits Competition 2021

2nd Place, E-Yantra International Robotics Competition 2021

Talks

"Thought-Adaptive KV Cache Compression for Efficient Reasoning Models"

NVIDIA, Santa Clara, CA (Host: Marina Neseem) Dec 2025

"Outlier-Aware Quantization: ViTs, LLMs and SSMs"

Tambe Lab, Stanford, CA (Host: Dr. Thierry Tambe) Mar 2026
A*STAR, Singapore (Host: Dr. Himeshi De Silva) Jan 2026
Apple, Cupertino, CA (Host: Dr. Fernando Mujica) Aug 2025

"Designing Quantized Accelerators for AI"

Google TPU Team, Sunnyvale, CA (Host: Dr. Krishnakumar Nair) Jul 2025
NVIDIA, Santa Clara, CA (Host: Dr. Marina Neseem) Jul 2025
ACM Student Research Competition @ MICRO Nov 2024

"Learning the Hardware Accelerator Design Space through Unified Representations"

EuroSys-ASPLOS 2025 Mar 2025

"Logarithmic-Posit Encodings"

IBM Research, New York, NY Jun 2024
Lemurian Labs, Menlo Park, CA (Host: Dr. Theodore Omtzigt) May 2024

"Deep Learning Model Quantization"

NVIDIA, Santa Clara, CA (Host: Dr. Visu Subramanian) May 2024
CoCoSyS Liaison Meeting, Online (Host: Emily Watson) Mar 2024

Media Coverage

Neptune.AI: Impressive research on efficient foundation model inference 2025

OpenVINO: Q2'25: Technology Update – Notable Research – ThinKV 2025

OpenVINO: Q4'25: Technology Update – Notable Research – OuroMamba 2025

Georgia Tech: Ramachandran won second place at the ACM Student Research Competition 2025

CoCoSyS: Director's Note: CoCoSyS Top-5 Accomplishments in Year 2, Quarter 3 2025

Georgia Tech: Research conference spotlight, International Conference on Computer Vision 2025

ACM SIGARCH: MICRO 2024 Trip Report: Success at Scale 2024

Grants

Research Program (Student Contributor)
DARPA Semiconductor Research Corporation (SRC) JUMP 2.0 CoCoSys Center (2025–2026)

Student Travel Grants
ICLR 2026, ISCA 2025, MICRO 2025, and DAC 2024; total support of approximately $6,000

Academic Service

Submission Co-Chair
MICRO 2026

Technical Program Committee Member (Conferences and Journals)
CoNGA 2024, ECCV 2024, SCOPE-ICLR 2025, IEEE COINS 2025, IEEE TCAD, ICCV 2025, NeurIPS 2025, CoNGA 2025, ICLR 2026, MLSys 2026, IEEE CAL, IEEE TIP, ICML 2026, BMVC 2026, NeurIPS 2026, IEEE TCSVT

Artifact Evaluation Committee
HPCA 2024, ASPLOS 2025, HPCA 2026, ISCA 2026

Teaching Experience

Hardware for Machine Learning (GT ECE 8803) Spring 2026
Guest Lecturer on Efficient AI, Prof. Divya Mahajan

Hardware for Machine Learning (GT ECE 8803) Spring 2025
Guest Lecturer on AI for Hardware, Prof. Tushar Krishna

Select Student Advisees

Matthew Nichols 2026–Present
Undergraduate intern from GT, USA

Daniel Moon 2026–Present
Undergraduate intern from SNU, South Korea

Mingyu Lee 2024–2026
GT undergraduate visiting student from KAIST, South Korea
Next destination: GT PhD Fall 2026

Pradyut Ganesh 2025
Undergraduate student from BITS Pilani, India
Next destination: CMU MS Spring 2026

References

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